

Homework 5: Differential Equations & Dynamics

Calculus II · STM 1002

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Due: Friday, June 5, 2026, at 11:30 am. **Submit as a hard copy in class.**

Problem 1: Verify Three Solutions

For each pair below, verify whether the candidate function solves the given DE. Differentiate, substitute, and confirm the equation holds for all t . Where an initial condition is given, also check it.

- (a) DE: $y' = 2y$, candidate $y(t) = 5e^{2t}$, with IC $y(0) = 5$.
- (b) DE: $y' = 1 + y^2$, candidate $y(t) = \tan t$, with IC $y(0) = 0$. (For what range of t is the candidate actually defined?)
- (c) DE: $T'(t) = -0.05(T - 22)$, candidate $T(t) = 22 + 68e^{-0.05t}$, with IC $T(0) = 90$. Also state the long-time limit $\lim_{t \rightarrow \infty} T(t)$.

Problem 2: Solve Four Separable IVPs

Solve each initial-value problem by separation of variables. Show the separated form and the integration step explicitly. Express y as an explicit function of the independent variable wherever possible.

- (a) $\frac{dy}{dt} = -3y$, $y(0) = 4$.
- (b) $\frac{dy}{dt} = ty$, $y(0) = 2$.
- (c) $\frac{dy}{dt} = y^2$, $y(0) = 1$. (Then state the value of t where your solution blows up.)
- (d) $\frac{dy}{dt} = 2(5 - y)$, $y(0) = 1$. (Hint: try the substitution $u = y - 5$, or separate directly.)

Problem 3: Fixed Points and Stability

For each autonomous DE below, do the following:

- Find all fixed points by solving $f(y) = 0$.
- Classify each as stable or unstable. State which test you used (phase line or linearization).
- Predict the long-time behavior of the solution starting from $y(0) = 2$.

- (a) $y' = 3y - 12$.

- (b) $y' = y(y - 4)$.
- (c) $y' = (y - 1)(y - 3)(y - 5)$.
- (d) $y' = \sin y$.

Problem 4: Coffee Shop: Find the Cooling Constant From Data

A new coffee shop wants to measure how fast its to-go cups cool, then use the model to make claims to customers. Coffee leaves the shop at temperature T_0 . Assume the cafe, outdoors, and the office are all approximately the same temperature, $T_a = 22^\circ\text{C}$ (72°F). The temperature obeys Newton's law of cooling,

$$\frac{dT}{dt} = -k(T - T_a),$$

with cooling constant $k > 0$ (units: min^{-1}).

- (a) **General solution:** Solve the DE for $T(t)$ in terms of T_0 , T_a , and k .
- (b) **Find k from a measurement:** An employee fills a cup at $T_0 = 85^\circ\text{C}$ (185°F) and records the temperature 15 minutes later as $T(15) = 50^\circ\text{C}$ (122°F). Use this datum to compute k . Round to three significant figures.
- (c) **Predict:** With your value of k from (b), predict the temperature 30 minutes after pickup.
- (d) **Design backwards:** The shop wants to advertise that coffee is “still hot” (defined as $T \geq 60^\circ\text{C}$ (140°F)) when the customer arrives at the office, 15 minutes after pickup. Using the same k , what is the minimum T_0 that meets this guarantee?

Problem 5: Drug Clearance and Dosing Schedule

A patient takes a 500 mg dose of a drug. The dose reaches peak plasma concentration $C_0 = 12 \mu\text{g/mL}$ almost immediately, then clears according to first-order pharmacokinetics with rate constant $k = 0.16 \text{ h}^{-1}$:

$$\frac{dC}{dt} = -kC.$$

- (a) Write down $C(t)$ from a single dose.
- (b) Find the half-life of the drug (the time for the concentration to drop to half of any starting value).
- (c) The minimum effective concentration is $4 \mu\text{g/mL}$. How long does a single 500 mg dose remain effective?
- (d) The patient now takes a 500 mg dose every τ hours, starting at $t = 0$. Show that the steady-state peak concentration (just after each dose) satisfies

$$C_{\text{peak}} = \frac{C_0}{1 - e^{-k\tau}}.$$

Hint. Just before the n th dose, the plasma concentration is whatever the previous peak decayed to over time τ . The n th dose then bumps it back up by C_0 . Write $C_{\text{peak}}^{(n)}$ in terms of $C_{\text{peak}}^{(n-1)}$, iterate, and recognize the geometric series.

- (e) The toxic threshold is $40 \mu\text{g/mL}$. What is the minimum dosing interval τ that keeps the steady-state peak under the threshold?

Problem 6: Deer Population With Hunting

A wildlife biologist models a deer population in a 10,000-acre preserve. Without hunting, growth is logistic:

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K} \right), \quad r = 0.4 \text{ year}^{-1}, \quad K = 1500 \text{ deer}.$$

Each year, hunters take a constant number h of deer:

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K} \right) - h.$$

- For $h = 0$, find all fixed points and classify each. What is the long-term population?
- For $h = 100$, find both fixed points (round to the nearest whole deer). Classify each as stable or unstable, and confirm with the linearization test.
- With $h = 100$, the herd starts at 800 deer. What does the population approach as $t \rightarrow \infty$? What if the herd starts at 200 deer? Explain in terms of the fixed points from (b).
- Compute the maximum sustainable harvest h_{max} — the value of h above which no fixed points exist. State the answer in deer per year, and explain in two or three sentences why a careful manager would set the actual harvest meaningfully below h_{max} .